

Article

Risk-Driven Sensor Placement in Sewer Networks: A Descriptive–Predictive–Prescriptive Framework

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Abstract

Sanitary sewer collection systems are among the least observable urban infrastructure assets, with most utilities operating fewer than one sensor per several hundred pipes; placement drives operational value. We develop DPP-SP, a Descriptive–Predictive–Prescriptive Sensor-Placement framework that links machine-learning failure prediction with risk-weighted maximum-coverage placement and apply it to a 33,349-pipe sewer system. The geographic information system (GIS) topology is rebuilt, raising the largest connected component from 29.8% to 89.4% of nodes. Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and a multilayer perceptron (MLP) are trained on combined 2020–2025 failure data; RF achieves the highest receiver-operating-characteristic area under the curve (ROC-AUC) of 0.7626 and supplies per-pipe risk weights, while repeated stratified cross-validation confirms this model ranking and permutation-importance and SHAP analyses identify pipe age and length as the dominant risk drivers. A budgeted maximum weighted coverage problem is solved over 680 candidate sites using greedy, genetic algorithm (GA) and tabu search (TS). At $K = 48$, RF with greedy covers 32.26% of network risk against an 11.73% baseline, a 174.9% improvement; all three optimizers converge on the same solution. Extending to $K = 400$ exposes a 56.58% coverage ceiling—set jointly by residual network fragmentation and the upstream detection range, and specific to the baseline candidate set and radius—and a six-radius sensitivity study (200–2500 m) identifies detection range as the most influential design parameter over the ranges tested. Risk coverage can be nearly tripled by redeploying the existing 48 stations without purchasing additional sensors.

Keywords: sensor placement; sewer monitoring; machine learning; submodular optimization; risk-based asset management; sanitary sewer overflow; smart water systems



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1. Introduction

Wastewater collection systems are among the largest and oldest assets that public utilities own. They are also among the least observable. The U.S. Environmental Protection Agency has estimated that between 23,000 and 75,000 sanitary sewer overflow events occur in the United States every year [1], and the American Society of Civil Engineers graded the country's wastewater infrastructure D+ in its most recent national report card, citing a renewal investment gap of more than USD 271 billion [2]. Continuous in-sewer monitoring with flow meters and level transmitters is the most direct way to detect anomalies early, but each installation is expensive to power, install, and maintain. As a result, most utilities run between a few dozen and a few hundred permanent stations across networks made up of tens of thousands of pipe segments. A recent industry survey of North American

water and wastewater utilities [3] confirms that condition-monitoring deployment remains uneven and is rarely targeted by formal risk criteria, and a companion overview of sensor-network optimization for smart wastewater and stormwater collection systems [4] frames the present study within the broader smart-infrastructure agenda. The utility that motivates this study currently runs 48 active stations across 33,349 pipes, a ratio of one sensor for every 695 pipes. Under that constraint, the choice of where a sensor goes effectively decides how much value the monitoring program delivers.

This work addresses a long-standing gap between two mature but separately developed research areas: sensor placement optimization in water and sewer networks, and machine-learning (ML) prediction of pipe failure. We propose a Descriptive–Predictive–Prescriptive Sensor-Placement (DPP-SP) workflow [5] in which an ML failure model produces per-pipe risk scores, and those scores then enter, as asset weights, a risk-weighted maximum-coverage placement problem. The full pipeline is built and applied to a 33,349-pipe sewer system operated by a large U.S. utility. Four questions guide the analysis. How much of the network’s risk do the existing 48 sensors observe, and how much could an optimized placement of the same 48 sensors observe? Which combination of ML model and optimizer produces the best risk coverage, and does the choice of optimizer matter at all? What is the marginal value of adding more sensors past the current budget, and is there a structural ceiling? Finally, how sensitive are the conclusions to the assumed upstream detection radius?

1.1. Related Work

Sensor placement in water distribution networks has a long history. Greedy submodular formulations [6,7], evolutionary methods, information-theoretic objectives [8] and integer-programming benchmarks [9] have all been studied. Krause et al. [6] showed that, because the coverage objective is submodular, a simple greedy selection is near-optimal, even at a large scale. Extensions to sewer systems are more recent. Banik et al. [10,11] compared five placement formulations on a 1902-node Italian sewer system and reported that greedy gave the best trade-off between solution quality and runtime. Wang et al. [12] used NSGA-II on a 4394-node network in Hong Kong, improving on centrality heuristics by 102%; NSGA-II is the non-dominated sorting genetic algorithm II. Wu et al. [13] combined spectral clustering with a joint-entropy objective for inflow and infiltration detection, outperforming the non-dominated sorting genetic algorithm III (NSGA-III) by 7 to 15%. Other contributions cover wastewater-based epidemiology [14], stormwater sparse sensing [15], real-time control [16,17], and digital twins [18]. Bakhshipour et al. [19] further demonstrated that multi-objective optimization of urban drainage layouts can deliver substantial resilience gains without raising capital cost. In every case, however, the coverage objective treats each upstream pipe as equally important.

In parallel, ML has been used extensively to predict pipe failure from physical attributes. Random Forest [20] has become the dominant algorithm; comprehensive reviews of sewer-pipe condition models identify tree-ensemble methods, and RF in particular, as the most consistently top-performing classifiers across utility datasets [21–23], while a recent bibliometric meta-analysis confirms age, material, and diameter as the most predictive features in the deterioration literature [24]. Laakso et al. [25] settled on a standard feature set with diameter, length, material, age, and gradient. Tavakoli et al. [26] and Hassouna et al. [27] confirmed their predictive value on real utility data. XGBoost [28] and neural networks [29] have also been applied, and class-imbalance techniques, such as under-sampling, the synthetic minority over-sampling technique (SMOTE), and K-means resampling, are well-benchmarked [30–32]. A recent study by Aounali et al. [33] added dynamic hydraulic features to Random Forest and raised ROC-AUC from the 0.60 to

0.68 range typical of static-attribute models up to 0.81 to 0.90. Across all this work, however, predicted failure probabilities are used only after prediction, as ranked lists for inspection or rehabilitation, and not as inputs to a placement optimizer.

Risk-based asset management treats risk as the product of failure likelihood and consequence [34] and has been used widely to prioritize sewer-pipe rehabilitation [35]. In water distribution networks, placement objectives that weight events by their probability and consequence are well-established: Leskovec et al. [7] formulated placement as maximizing an expected penalty reduction weighted by event probability and impact and showed that weighting changes the resulting design—optimizing the population affected concentration sensors in high-consequence areas, whereas treating all events as equally likely spread them uniformly. The range of such objective metrics was surveyed by Rath et al. [36]. Weighing coverage by the predicted failure likelihood of individual assets has not, to our knowledge, been delivered operationally for sewer collection systems. This study closes that gap: each pipe's ML-derived risk score becomes the asset weight in a maximum weighted set-cover formulation, and the resulting pipeline is the first end-to-end DPP workflow of this kind reported for sewer systems.

1.2. Research Gap

Despite this substantial body of work, three gaps remain. First, sewer sensor-placement studies almost universally adopt uniform or geometric coverage objectives, in which every pipe is weighted equally, rather than weighting coverage by each pipe's predicted failure risk; probability- and consequence-weighted objectives are established in water distribution networks [7,36] but has not, to our knowledge, been delivered operationally for sewer collection systems. Second, machine-learning failure prediction and placement optimization have developed through largely separate research efforts, and no end-to-end predictive-to-prescriptive pipeline that feeds ML-derived pipe risk scores directly into a coverage optimizer has been reported and validated on a real sewer network. Third, existing placement studies are typically demonstrated on synthetic or idealized networks and benchmarked against heuristic baselines, rather than against a real utility's actual in-field sensor layout, and they rarely quantify the structural observability limits that the real GIS topology imposes on achievable coverage. This paper addresses all three gaps directly, as summarized in the contributions below.

The contributions of the paper are as follows. First, we replace the uniform-coverage assumption used in earlier sewer placement studies with a risk-weighted objective. Second, we deliver an end-to-end ML-to-optimization pipeline that operationalizes the predictive-to-prescriptive coupling identified as a gap in the literature. Third, we report a six-radius parametric sensitivity study that quantifies the influence of the detection-range assumption. Fourth, we benchmark the optimized configurations against the existing sensor layout of a real utility rather than against a synthetic baseline. Fifth, we identify and quantify a structural observability ceiling caused by residual isolated fragments. Sixth, we use a fixed-denominator coverage definition so that results from different ML models can be compared directly. The rest of the paper is organized as follows. Section 2 introduces the study area, the datasets, and the methodology. Section 3 reports coverage of the current budget, $K = 1$ to 400 sensitivity, and radius sensitivity. Section 4 discusses the results and their implications. Section 5 concludes the paper.

2. Materials and Methods

2.1. Study Area and Data

The framework is applied to the sanitary sewer collection system of a large, rapidly growing U.S. County with a current service population above 450,000. The utility name is

withheld under a non-disclosure agreement and is referred to throughout as ‘the utility’. The case-study utility is anonymized in accordance with data confidentiality requirements; all maps, geographic coordinates, asset identifiers, and utility-identifying information were removed or transformed prior to reporting. The network spans about 1917 km (1191 mi) of pipes and is made up of gravity collection mains, pressurized force mains, pump (lift) stations, and meter vaults. All spatial data are stored in a U.S. survey-foot projected coordinate system; the specific projection is withheld to avoid identifying the case-study utility.

Five categories of data were combined. Pipe segment geometries and attributes were pulled from the utility’s geographic information system (GIS) database: 32,039 gravity mains, 1310 force mains, 31,865 sewer manholes, 34 pump stations, 77 large meter vaults and 127 sewer vaults. Each segment carries nominal diameter, material, installation year, slope, and plan length. Three independent failure data sets covering 2020 to 2025 supplied the binary failure labels: closed-circuit television (CCTV) pipe-inspection records (2025, collected by the utility using ITpipes inspection software, <https://itpipes.com/inspection-software/> (accessed on 28 May 2026)), the utility’s sewer-issues log of service disruptions, and the geo-referenced sanitary sewer overflow history. Real-time flow and level data from the 48 instrumented sites (26 sewer-vault meters and 22 pump-station flow monitors) provided the baseline against which the optimized placements are compared.

The attribute completeness and currency of these inputs are high, which matters because the risk model depends on them. In the analytical dataset, nominal diameter and slope are populated for 100% of pipes, material for 99.9%, and installation year for 99.2%; invert-in, invert-out, and the derived elevation drop are available for every segment. Pipe age was computed as the difference between the 2025 reference year and the installation year, giving a median age of 21 years (0 to 63 years after clipping); the small fraction of segments carrying a placeholder installation date were assigned the network median and their age clipped, as described in Section 2.3. The failure labels are drawn from current records—ITpipes CCTV inspections from 2025 together with maintenance and sanitary sewer overflow histories spanning 2020 to 2025—and 22,130 pipes (66% of the network) carry at least one CCTV observation, so the training signal is both recent and spatially representative of the service area rather than confined to a small sample. Because the GIS geodatabase is the utility’s live enterprise asset register, the diameter, elevation, and age attributes reflect the current recorded state of the field network; residual attribute error is bounded by the completeness figures above and is further mitigated by the annual retraining cycle recommended in Section 4.

The monitoring devices considered in this study are in-sewer hydraulic sensors—flow meters and level (depth) transmitters—rather than water-quality instruments. The utility’s 48 existing stations comprise 26 sewer-vault flow meters and 22 pump-station flow and wet-well level monitors; each reports a continuous flow rate and water level, the quantities used operationally for hydraulic monitoring (flow accounting, wet-well control, and overflow detection). For the placement problem, a sensor is treated abstractly as a device that observes the hydraulic state of the pipes upstream of its location within the detection range, so that a developing blockage, surcharge, or structural failure that perturbs flow or depth upstream can be detected. The framework is therefore agnostic to the specific make or model of sensor and applies to any in-pipe flow- or level-sensing technology the utility elects to deploy; it does not require, and the analysis does not assume, water-quality sensing.

The raw GIS extract contained 130 disconnected components. The largest covered only 9493 nodes, or 29.8% of the network, for three reasons: incomplete manhole records at pipe-to-pipe junctions, vault structures stored in separate CSV files, and 92% of force

mains that could not snap to any node. A further 28 pairs of duplicate-coordinate nodes were also identified. We performed a four-step additive topology rebuild: 200 vault nodes were integrated; 1407 virtual junction nodes were created at clusters of coincident pipe endpoints; all 33,349 pipe endpoints were re-snapped with a k-dimensional (KD) tree at a 15 m (50 ft) tolerance, taking snap success from 95.7% to 99.5%; and the 28 duplicate pairs were merged. The reconstructed network has 33,474 nodes and 33,349 directed edges in 78 connected components, with the largest covering 29,914 nodes (89.4%). The largest component therefore grew by 59.6 percentage points (Table 1; Figure 1).

Table 1. Network topology in the raw GIS extract compared to the reconstructed analytical network.

Metric	Raw GIS Extract	Reconstructed Network
Total nodes	31,894	33,474
Virtual junction nodes	—	1407
Vault nodes	—	200
Snap success rate	95.7%	99.5%
Force-main snap rate	~8%	89.2%
Connected components	130	78
Largest component (nodes)	9493 (29.8%)	29,914 (89.4%)

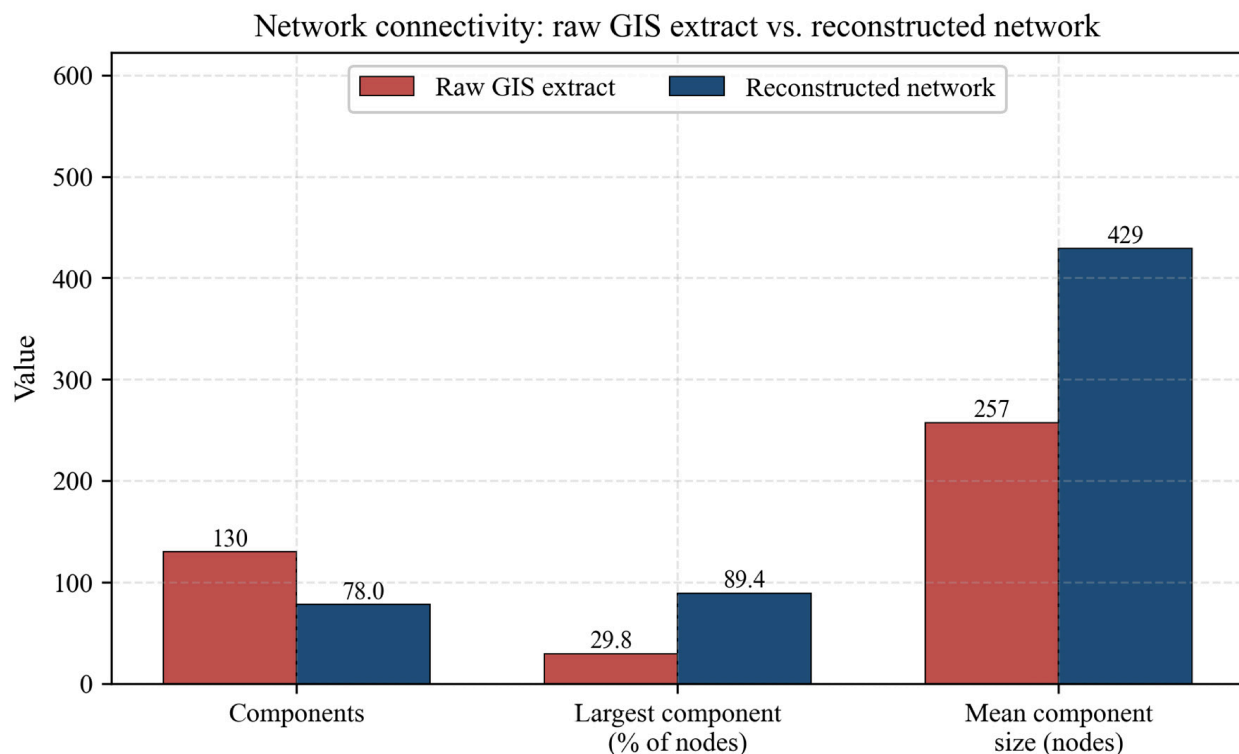


Figure 1. Network connectivity in the raw GIS extract compared to the reconstructed analytical network. Adding 200 vault nodes and 1407 virtual junctions, plus re-snapping all pipe endpoints with a 15 m (50 ft) tolerance, reduced the number of disconnected components from 130 to 78 and grew the largest component from 29.8% to 89.4% of all nodes.

The rebuild is strictly additive and fully traceable, so the reconstructed graph augments rather than replace the field network. No original manhole, pipe, or pump station was deleted, moved, or geometrically altered; the procedure only inserts the missing connective elements (the 200 vault nodes and 1407 virtual junctions) and updates connectivity flags.

Every node carries a type of label—manhole, pump station, meter or sewer vault, or virtual junction—so the reconstructed elements remain demarcated from the original assets and can be filtered or audited at any time against the source GIS. The re-snapping step likewise preserves pipe geometry: each pipe endpoint is associated with its nearest existing node within the 15 m (50 ft) tolerance without changing the pipe's coordinates or length. The analytical network is therefore a connectivity-complete superset of the utility's recorded network, in which the added elements restore the hydraulic continuity that the raw export had lost to data-entry gaps while leaving every original field asset intact and identifiable.

For clarity, the terms used in Table 1 are defined as follows. Vault nodes are underground meter and access structures—large meter vaults and sewer vaults—that house flow meters or provide network access; in the raw extract these were stored in separate files and were absent from the pipe graph, so 200 of them were added as nodes during the rebuild. Virtual junction nodes are connection points inserted where two or more pipe endpoints share the same coordinates, but no manhole record exists; 1407 were created so that pipes meeting at these unrecorded junctions became topologically connected. The snap success rate is the percentage of pipe endpoints successfully matched ('snapped') to a network node within the 15 m (50 ft) search tolerance—a direct measure of how completely the pipes join into a connected graph—which the rebuild raised from 95.7% to 99.5%. The force-main snap rate is the same measure restricted to pressurized force mains, which are stored in a separate GIS layer and are the hardest to connect; it rose from about 8% to 89.2%. Connected components and largest components then summarize the resulting connectivity: the number of separate sub-networks, and the share of all nodes contained in the single largest one. The reconstructed node total (33,474) equals the 31,894 raw nodes plus 1407 virtual junctions and 200 vault nodes, net of 27 merged coincident-coordinate nodes (identified among the 28 duplicate pairs). The raw node count falls just below the manhole-plus-pump-station inventory of Section 2.1 (31,899) because coincident and invalid-coordinate points are dropped during graph construction, and 200 of the 204 cataloged vaults had usable coordinates for insertion.

Beyond network topology, the condition signal used for training spans a broad range of sewer defect types drawn from the three failure sources introduced above. Structural defects recorded in the CCTV inspections include surface damage and corrosion (spalling and exposed aggregate or reinforcement); circumferential, longitudinal, and multiple cracks; fractures; breaks and holes; and, in the most severe cases, deformation and collapse. Operation-and-maintenance and hydraulic defects include root intrusion, joint and barrel infiltration, and settled or attached deposits that obstruct flow (blockages), together with sag or reverse-slope conditions inferred from standing-water levels. Each observation is graded on the NASSCO Pipeline Assessment Certification Program (PACP) 1-to-5 severity scale; Figure 2 summarizes the composition of the graded observations. The maintenance work-order log adds service disruptions such as backups and blockages, and the sanitary sewer overflow history records the ultimate hydraulic failure outcome. A pipe receives a positive failure label when any of three conditions holds: a maximum PACP structural grade of 3 or higher in the CCTV records, at least one reported sewer-maintenance issue, or at least one recorded sanitary sewer overflow event; this union rule yields the 485 failure-labeled pipes (1.454%) used for training, so only a fraction of the CCTV-inspected pipes—those with a severe maximum grade or an associated failure record—are labeled failures. Because the binary failure label is formed from the union of these sources, the model targets a general notion of pipe distress that integrates structural deterioration and flow-obstruction mechanisms rather than any single defect type.

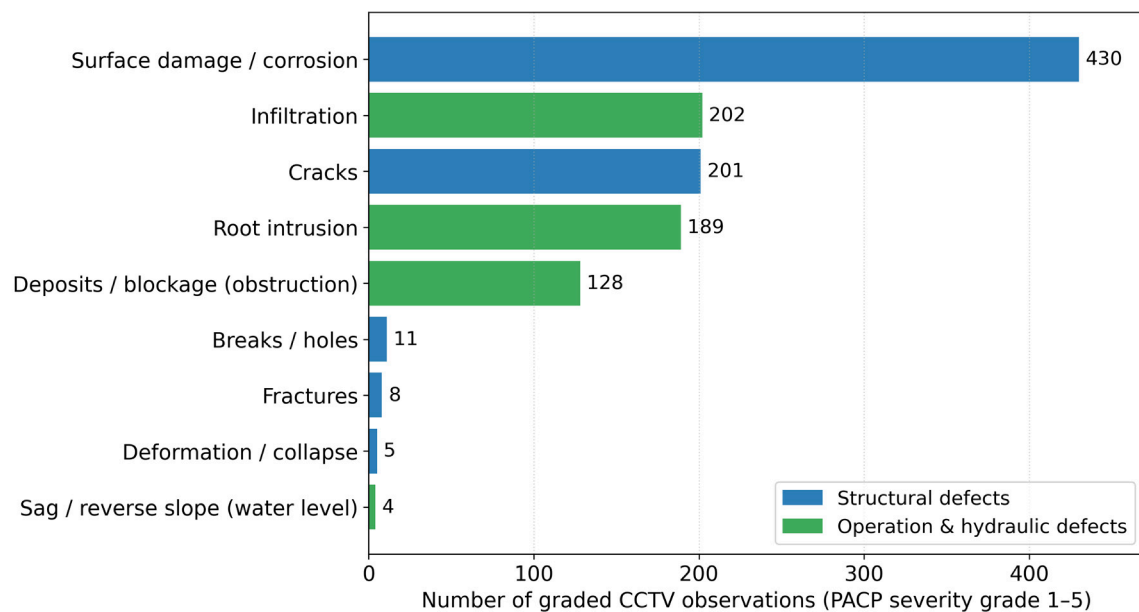


Figure 2. Composition of the graded (PACP severity 1–5) defect observations recorded in the CCTV inspection data, separated into structural defects (blue) and operation-and-hydraulic defects (green). The binary failure labels used for training are drawn from the union of these CCTV records, the maintenance work-order log, and the sanitary sewer overflow history. Figure 2 counts individual graded CCTV observations, not labeled pipes.

2.2. Problem Formulation

We let $G = (V, E)$ be the directed graph that represents the sewer network, with V the set of manholes and junctions and E the set of directed pipe segments oriented in the direction of gravity flow. We let A be the set of $n = 33,349$ monitored assets ($A \subset E$) and C be the set of $m = 680$ feasible candidate sensor locations drawn from pump stations, vaults, and high-confluence manholes ($C \subset V$). For each candidate $j \in C$, the upstream reachable set D_j collects every asset $i \in A$ whose graph distance $d(i, j)$ is at most the detection range $L_{\max} = 1500$ m (4921 ft). That range reflects the practical signal-propagation window of typical in-sewer flow and quality instruments. At gravity-sewer design velocities of 0.6 to 1.2 m/s [37,38], 1500 m corresponds to an upstream travel time of 21 to 42 min. D_j is computed by reverse Dijkstra: edge directions are reversed and a single-source shortest-path search is run from j . The placement problem is then selecting a subset $S \subset C$ of exactly K locations that maximizes total risk-weighted coverage. This is a maximum weighted set-cover problem, which is NP-hard in general [39]. The coverage objective is, however, monotone submodular, so the greedy algorithm is guaranteed to achieve at least $(1 - 1/e)$ (about 63%) of the optimal coverage [40].

2.3. Failure-Probability Estimation

The prediction stage deliberately relies on established, extensively benchmarked classifiers rather than a novel learning algorithm. This is a considered design choice: the contribution of this work is not a new failure-prediction model but the end-to-end coupling of an ML-derived pipe-risk surface to a risk-weighted placement optimizer. Using mature, well-understood methods (Random Forest, XGBoost, and a multilayer perceptron) maximizes reproducibility and lowers the barrier to adoption by utilities, which must be able to trust, audit, and re-run the model on their own data with widely available, open-source tools. The three classifiers are selected to span the model-complexity spectrum—a bagged tree ensemble, a boosted tree ensemble, and a neural network—so

that the sensitivity of the downstream placement to the choice of predictor can itself be quantified rather than assumed.

Each model was selected for a specific reason. Random Forest is the primary model and the one retained for deployment, as tree ensembles, and Random Forest in particular, are repeatedly identified in sewer-condition reviews as the most consistently top-performing classifiers on utility data [21–23]; the method handles mixed categorical and numeric attributes and non-linear feature interactions without extensive preprocessing, is comparatively robust to overfitting and to hyperparameter choice, and yields the feature-importance and SHAP attributions reported in Section 3.4. XGBoost is included to represent gradient-boosted trees—a family that frequently achieves state-of-the-art accuracy on tabular data—and provides a stringent test of whether boosting improves on Random Forest’s bagging for this problem. The multilayer perceptron represents the neural-network family and tests whether a connectionist model adds value on a static, eight-feature tabular task of this size. All three natively output a continuous failure probability—required to form the per-pipe risk ranking that drives the optimizer—accommodate class-imbalance weighting and are available in mature open-source libraries. Deep-learning and graph-neural approaches were not pursued because the predictors are static per-pipe attributes with a modest positive-class count, a regime in which tree ensembles are the established best choice; incorporating dynamic hydraulic features, which have been shown to raise predictive accuracy [33], is identified as future work.

Three classifiers were trained to estimate the per-pipe failure probability p_i . Each pipe is represented by an eight-feature vector: diameter, age (computed as 2025 minus install year and clipped to the range 0 to 100), slope, length, upstream node degree, downstream node degree, and label-encoded material and pipe class; diameter, age, slope, length, material, and pipe class are read directly from the GIS attribute tables, whereas only the two node-degree features are computed from the reconstructed graph topology. Binary labels ($y_i = 1$) were assigned by spatial join of the three-failure datasets to the pipe geometries; 485 of the 33,349 pipes were labeled as failures, a 1.454% failure rate. We used a stratified 75/25 train–test split. The three models compared are Random Forest (300 trees, balanced class weights), XGBoost with scale_pos_weight, and a multilayer perceptron (MLP, two hidden layers of 128 and 64 units, rectified linear unit (ReLU) activations, dropout 0.3, class-weighted cross-entropy). All models were implemented in Python 3.11 using scikit-learn 1.4.2 (Random Forest, MLP) and XGBoost 2.0.3. The models were compared on the 8338-sample holdout using the receiver-operating-characteristic area under the curve (ROC-AUC), the precision–recall area under the curve (PR-AUC) and the F1-score, defined as the harmonic mean of precision and recall (Table 2). The best model by ROC-AUC was then retrained onto the full dataset for deployment. Figure 3 visualizes the same comparison and highlights the divergence between ROC-AUC and F1 across the three models.

Table 2. Machine-learning model performance on the 25% stratified holdout test set (8338 pipes with 121 failures, drawn from the full dataset of 33,349 pipes and 485 failures). ROC-AUC = receiver-operating-characteristic area under the curve; PR-AUC = precision–recall area under the curve; F1-score = harmonic mean of precision and recall.

Model	ROC-AUC	PR-AUC	F1-Score
Random Forest	0.7626	0.0844	0.1684
XGBoost	0.7323	0.0654	0.1134
Neural Network (MLP)	0.6708	0.0800	0.1806

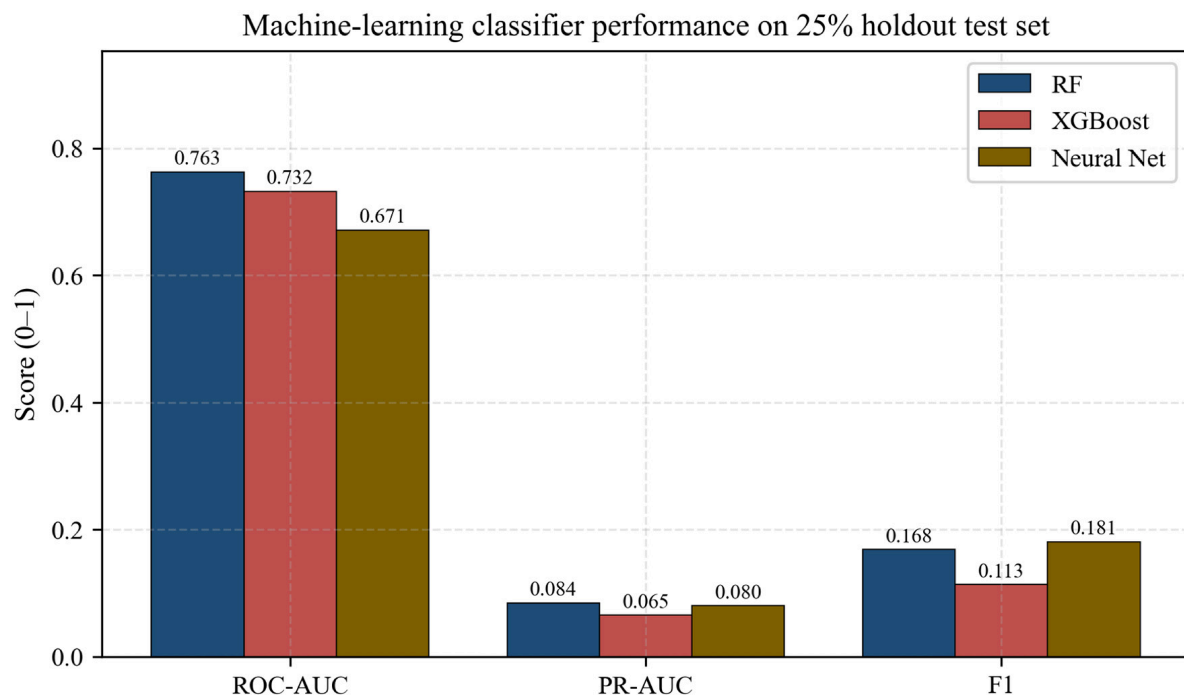


Figure 3. Comparison of the three ML classifiers on the 25% holdout. Random Forest leads on ROC-AUC (the ranking-quality metric most relevant to sensor placement); the MLP achieves the highest F1 but the lowest ROC-AUC, illustrating that classification accuracy and ranking quality are distinct criteria.

Because failures constitute only 1.454% of pipes, the class imbalance was handled at the model level rather than by resampling: Random Forest and the MLP used balanced (inverse-frequency) class weights, and XGBoost used `scale_pos_weight` set to the negative-to-positive ratio. Model hyperparameters were fixed a priori using the standard literature-informed values (Random Forest: 300 trees, maximum depth 15; XGBoost: maximum depth 6, learning rate 0.1; MLP: 128 and 64 hidden units) rather than tuned against the test set, so the reported holdout metrics are not optimistically biased by hyperparameter search.

To confirm that the model ranking is not an artifact of a single train–test partition, the comparison was repeated under five-times-repeated stratified five-fold cross-validation (25 folds). The ranking is stable and the separation between Random Forest and the other two classifiers is statistically robust, with non-overlapping 95% confidence intervals: Random Forest attains a mean ROC-AUC of 0.750 (95% CI 0.741–0.758), versus 0.714 (0.705–0.723) for XGBoost and 0.658 (0.647–0.670) for the MLP; the corresponding PR-AUC values are 0.098 (0.089–0.107), 0.077 (0.070–0.084) and 0.043 (0.039–0.047). The single-split holdout ROC-AUC reported in Table 2 (0.7626 for Random Forest) lies within one standard deviation of the cross-validated mean, indicating that the headline metrics in Table 2 are representative rather than a favorable draw.

2.4. Risk Score and Candidate Locations

The per-pipe risk score is defined as $r_i = p_i \cdot s_i$, combining the predicted failure probability with a consequence severity weight s_i that depends on the diameter and pipe type, following EPA and ASCE infrastructure risk-matrix conventions [2,41]. We use $s_i = 1$ for diameters below 203 mm (8 in.), $s_i = 2$ for diameters between 203 and 457 mm (8 to 18 in.), and $s_i = 3$ for diameters of 457 mm (18 in.) and above or for any force main. A key methodological choice is to use a fixed reference total $\text{REF}_{\text{TOTAL}} = \sum r_i = 11,311.44$ with risk units as the single, fixed risk surface against which every placement is scored, held constant across all three ML models. Both the risk captured by a placement and the

normalizing total are taken from this one reference surface, so every model is evaluated on the same yardstick, and the maximum attainable coverage is 100% for each, in principle, before the network-structure limits quantified in Section 3.2. Crucially, this reference surface is the fixed risk map produced by the full-data deployment pipeline and is distinct from the three comparison classifiers' own recomputed predicted risk totals (RF: 10,466.76; XGBoost: 7110.37; NN: 4745.76), which differ by more than a factor of two because the class-weighted models are uncalibrated and their probability scales are not directly comparable. Scoring each model against its own total would confound ranking quality with these arbitrary scale differences, while scoring all models against the one fixed reference surface removes that confound; therefore, the coverage percentages in Table 3 measure how much of a common risk surface each model's placement captures, and the differences between models reflect which high-risk assets each ranking selects rather than differences in absolute probability mass. Because greedy maximizes a sum of risk values, multiplying an entire risk vector by a positive constant leaves the selected set—and therefore its Table 3 coverage—unchanged, so the model comparison is unaffected by differences in overall probability scale; the ranking of the three models is thus governed by ordering quality rather than by total predicted mass, consistent with their held-out ROC-AUC values (0.7626, 0.7323, and 0.6708 for the Random Forest, XGBoost, and neural-network models, respectively).

Table 3. Coverage results for all sensor-placement methods ($K = 48$, $REF_{TOTAL} = 11,311.44$).

ML Model	Optimizer	Coverage%	Risk Covered	Assets Covered	Δ vs. Existing	Time (s)
—	Existing baseline	11.73	1327.19	3355	—	—
RF	Greedy	32.26	3649.11	8187	+174.9%	<1
RF	Genetic	32.26	3649.11	8187	+174.9%	11
RF	Tabu search	32.26	3649.11	8187	+174.9%	117
XGBoost	Greedy	31.70	3586.04	7733	+170.2%	<1
XGBoost	Genetic	31.70	3586.04	7733	+170.2%	8
XGBoost	Tabu search	31.70	3586.04	7733	+170.2%	94
NN (MLP)	Greedy	30.62	3463.64	7344	+161.0%	<1
NN (MLP)	Genetic	30.62	3463.64	7344	+161.0%	8
NN (MLP)	Tabu search	30.62	3463.64	7344	+161.0%	87

The 680 candidate locations C were drawn from existing sensor sites (48), pump stations and vault nodes, and high-confluence manholes with in-degree of at least 2, ranked by decreasing in-degree. All candidates were snapped to the nearest graph node and deduplicated; virtual junction nodes, which are GIS artifacts, were excluded.

2.5. Optimization Algorithms and Coverage Metric

Three algorithms were applied and compared. The greedy algorithm selects, at each of the K steps, the candidate that produces the largest marginal gain in covered risk. A genetic algorithm (GA) was run with a population of 60 over 100 generations, tournament selection, and a mutation rate of 0.02, and was seeded from the greedy solution. Tabu search (TS) was run for 300 iterations with two random restarts and a tabu tenure of 20,

again seeded by the greedy result. All methods are evaluated on the same fixed-reference coverage ratio, defined in Equation (1):

$$\text{Coverage}(\%) = \frac{\sum_{i \in \text{covered}} r_i}{\text{REF}_{\text{TOTAL}}} \times 100 \quad (1)$$

Because GA and TS are warm started from the greedy solution and configured for moderate exploration budgets typical of operational deployment (GA: POP = 60, GEN = 100; TS: 300 iterations $\times$ 2 restarts), they function as refinement and robustness checks on the greedy solution rather than as independent competitors. The near-identical results across all three optimizers reported in Section 3.1 are therefore consistent with the $(1 - 1/e)$ submodular guarantee: greedy is guaranteed to achieve at least 63% of the optimal value, and the local search performed by GA and TS confirms that no nearby swap-based improvement exists.

3. Results

3.1. Coverage at the Current Budget ($K = 48$)

Random Forest produced the highest ROC-AUC (0.7626), meaning it ranks pipes by failure probability better than the other two classifiers. Table 3 reports the coverage achieved by all nine combinations of the ML model and optimizer and the existing baseline, all measured against $\text{REF}_{\text{TOTAL}} = 11,311.44$. Every optimized method beats the baseline by a wide margin. The 48 sensors that are currently in the field cover only 11.73% of network risk, while all nine optimized combinations reach at least 30.62%, a minimum improvement of 161.0%. The best combination is Random Forest with greedy placement at 32.26% coverage (3649.11 risk units, 8187 assets), an improvement of 174.9% over the existing layout. Figure 4 displays the same comparison graphically.

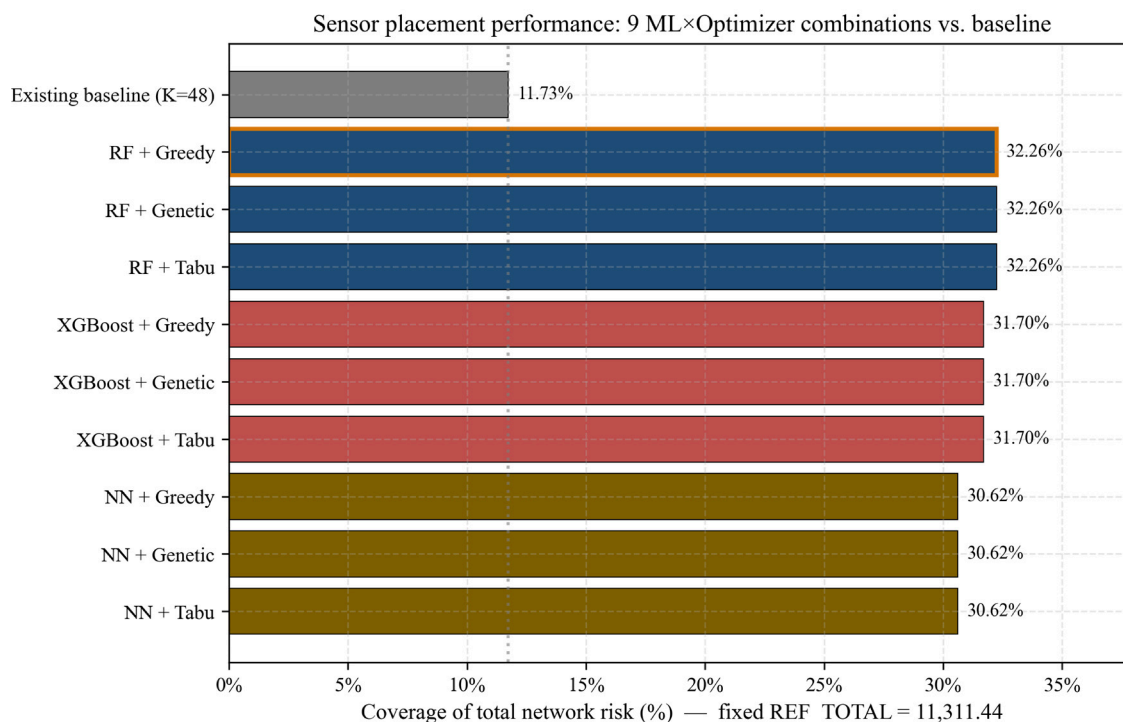


Figure 4. Risk coverage at $K = 48$ for every method tested, all against the fixed denominator $\text{REF}_{\text{TOTAL}} = 11,311.44$. All three optimizers tie within each ML model, isolating ML model choice

as the only meaningful performance lever at this problem scale. The orange-outlined bar marks the configuration adopted throughout this study (Random Forest risk scores with greedy placement, 32.26% coverage).

To clarify what the ‘risk covered’ column of Table 3 represents, for each pipe that falls within the upstream detection range of a placed sensor, it is the sum of the pipe’s risk on the single fixed reference surface introduced in Section 2.4, expressed in risk units and reported as a percentage of that same fixed reference total; because every ML model is scored against this one common surface, the coverage percentages are directly comparable. As defined in Section 2.4, each pipe’s risk score is the product of two components, in the classical risk sense of likelihood times consequence. The likelihood component is the machine-learning-predicted probability of failure, learned from the eight physical and topological attributes (diameter, age, slope, length, upstream and downstream node degree, material, and pipe class) against the 2020–2025 failure record of CCTV defects, maintenance work orders, and sanitary sewer overflows. The consequence component is a severity weight that increases with pipe diameter and is highest for large-diameter and pressurized force mains, following EPA and ASCE risk-matrix conventions, because failures on larger and pressurized mains carry greater service-disruption and environmental impacts. The ‘risk covered’ value therefore rewards a placement for observing pipes that are both likely to fail and costly if they do, rather than simply maximizing the number of pipes seen; the consequence dimensions not yet included—population served, proximity to sensitive receptors, and repair cost—are noted as a direction for refinement in the Limitations.

One result worth highlighting is that, for every ML model, the GA and TS solutions match the greedy solution to within the resolution of the coverage metric ($\Delta < 0.01$ percentage points). This is consistent with greedy being near-optimal for submodular coverage problems; with the $(1 - 1/e) \approx 0.63$ Nemhauser guarantee already in hand, the role of GA and TS here is to verify that no nearby swap-based improvement exists, not to outperform greedy as an independent competitor. The choice of ML model, on the other hand, does have a clear effect: RF beats XGBoost by 0.56 percentage points and the NN by 1.64 percentage points, because it ranks the high-risk pipes better. The takeaway is that greedy over the complete asset pool is the right formulation for this problem, and the agreement of GA and TS provides empirical confirmation of that choice.

The ‘assets covered’ column varies across the three ML models for a straightforward reason: the optimization maximizes risk-weighted coverage, not the number of pipes observed. Each classifier produces a different predicted-risk surface, so the greedy solver selects different candidate locations, and those locations reach different sets of upstream pipes. A model that concentrates predicted risk on a smaller number of large-consequence mains leads greedy to sensors that cover fewer but higher-value assets, whereas a model that spreads risk more evenly leads to placements that touch more pipes. This is why, at comparable coverage percentages, Random Forest covers 8187 assets, XGBoost 7733, and the neural network 7344: the asset count is a by-product of the sensor set each risk surface drives the optimizer to choose, not a quantity that is directly optimized. Because all three models are evaluated against the same fixed reference total, the coverage percentages remain directly comparable even though the underlying asset counts differ.

3.2. Marginal Sensor Value ($K = 1$ to 400)

To study how much each additional sensor contributes, the greedy algorithm with Random Forest risk scores ran for K from 1 to 400. Table 4 shows the main milestones. Coverage grows quickly early on, through the efficiency elbow near $K = 48$, and up to about $K = 78$ and then slows down sharply, settling at 56.58% by $K = 400$. This plateau reflects a structural coverage ceiling that is specific to the current candidate set

and the 1500 m detection radius. About 43% of the risk-weighted assets are unobservable under this configuration, either because they sit inside one of the 77 residual isolated fragments that contain no candidate sensor, or because they lie more than 4921 ft upstream of every candidate. The full coverage trajectory and per-sensor marginal gains are shown in Figures 5 and 6, respectively.

Table 4. Coverage milestones from greedy sensor placement (RF with greedy, $K = 1$ to 400, $REF_{TOTAL} = 11,311.44$).

Sensor Counts K	Coverage%	Δ vs. $K = 48$
43	30.17	−2.09 pp
48 (current)	32.26	—
50	32.90	+0.64 pp
78	40.12	+7.86 pp
100	43.87	+11.61 pp
150	49.54	+17.28 pp
155	49.98	+17.72 pp
200	52.75	+20.49 pp
250	54.57	+22.31 pp
300	55.72	+23.46 pp
350	56.34	+24.08 pp
400	56.58	+24.32 pp (plateau)

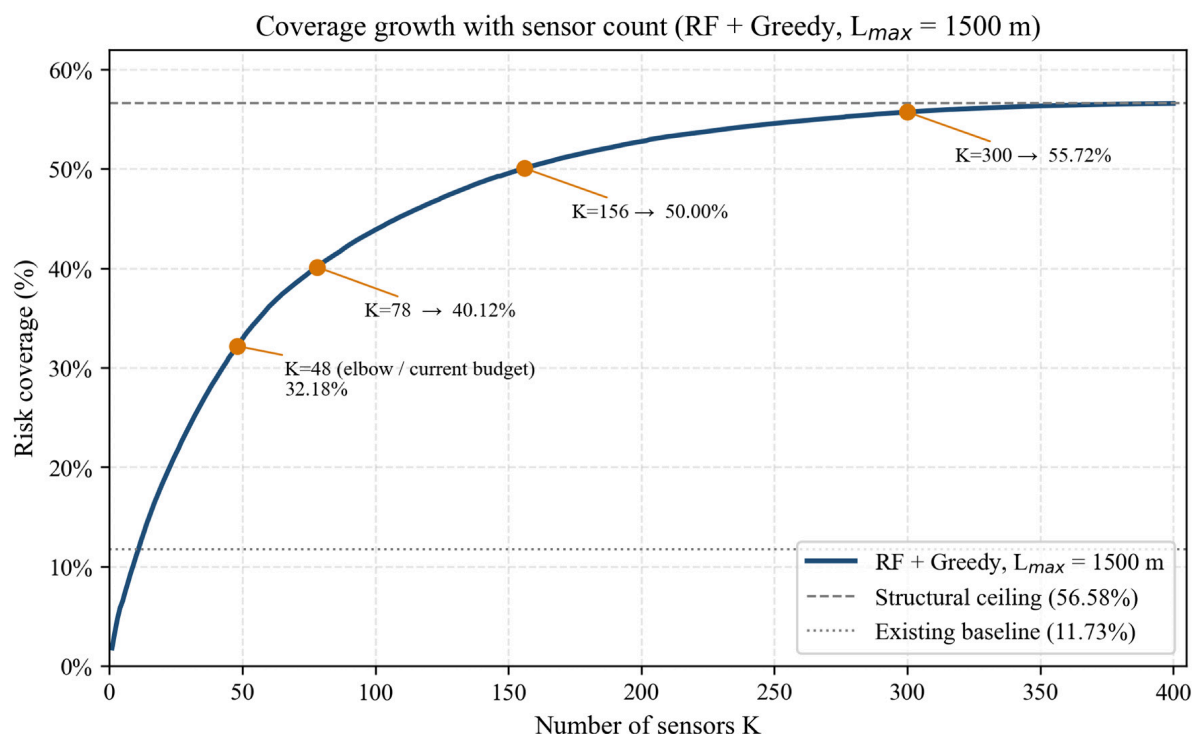


Figure 5. Greedy coverage growth from $K = 1$ to $K = 400$ (RF risk scores, $L_{max} = 1500$ m). The curve passes through the current budget ($K = 48$, 32.26%), the 40% milestone ($K = 78$), and the 50% milestone ($K = 156$), then asymptotes to the structural ceiling of 56.58% imposed by isolated network fragments and the upstream-distance constraint.

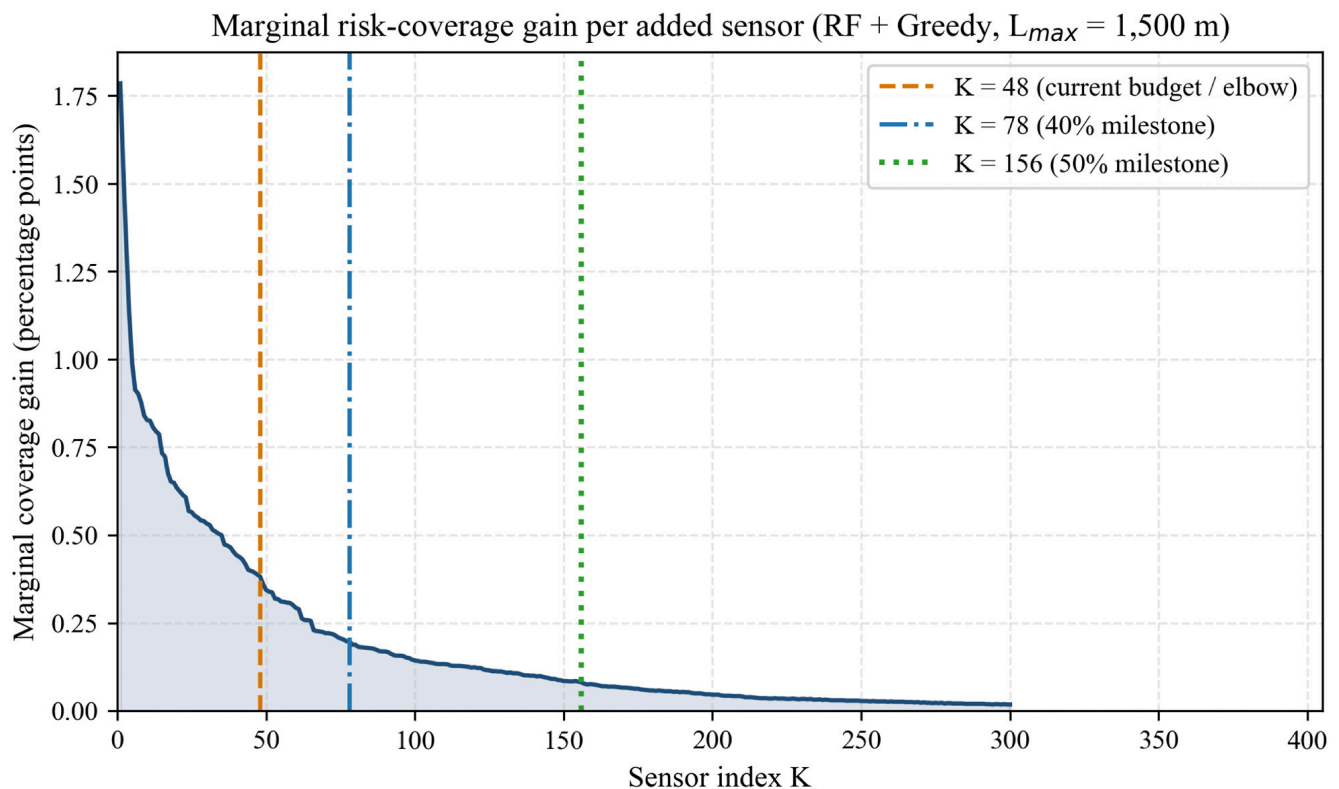


Figure 6. Marginal risk-coverage gain contributed by each additional sensor (RF + Greedy). Per-sensor returns drop steeply through the $K = 48$ elbow, fall below 0.4 pp sensor⁻¹ beyond $K = 156$, and become negligible past $K \approx 300$.

3.3. Detection-Radius Sensitivity

The detection radius $L_{\max}$ is a key design parameter, and its value depends on sensor technology, flow velocity, and signal attenuation. We re-ran the greedy algorithm with Random Forest risk scores from $K = 1$ to 400 for six values of $L_{\max}$ as 200, 500, 1000, 1500, 2000, and 2500 m, keeping every other parameter fixed. Coverage at $K = 48$ varies by a factor of ten across this range, from 4.22% at 200 m up to 43.30% at 2500 m, which, over the parameter ranges examined here, makes detection range more influential on coverage than either the ML model or the optimizer choice. The structural ceiling rises with radius, but it does not approach 100%; even at 2500 m, it only reaches 63.31% because isolated fragments impose an upper bound that no detection radius can overcome. The efficiency elbow shifts toward lower K at longer radii ($K = 71$ at 200 m, $K = 48$ at 1500 m, $K = 24$ at 2500 m); at very short radii, it first rises slightly (from 71 at 200 m to 79 at 500 m) before declining. Overall, extending detection range is, in effect, equivalent to buying more sensors. The full set of coverage curves and the corresponding ceiling/elbow summary are shown in Figures 7 and 8.

The spatial distribution of the per-pipe predicted risk scores $r_i = p_i \cdot s_i$ is shown in Figure 9 and reveals a strong spatial concentration of high-risk assets, consistent with the observed historical failure clusters, supporting the risk-weighted coverage objective adopted in this work.

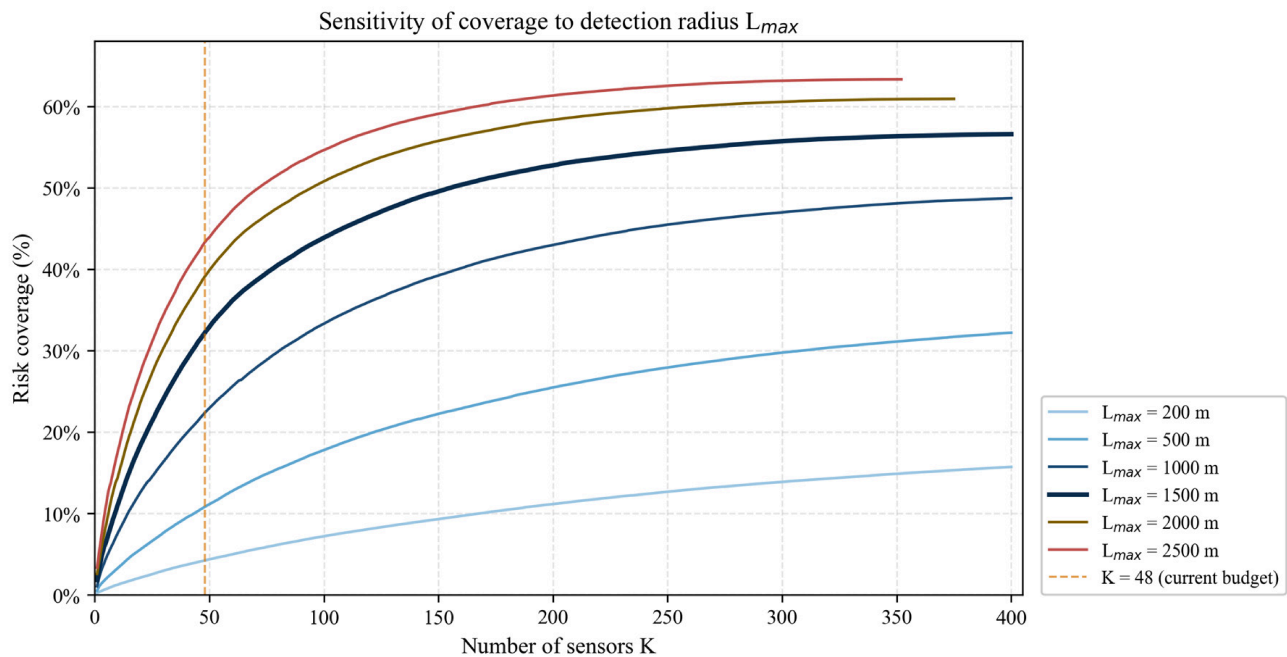


Figure 7. Greedy coverage curves for six detection radii $L_{\max} \in \{200, 500, 1000, 1500, 2000, 2500\}$ m. The 1500 m baseline is highlighted; the vertical dashed line marks the current sensor budget $K = 48$.

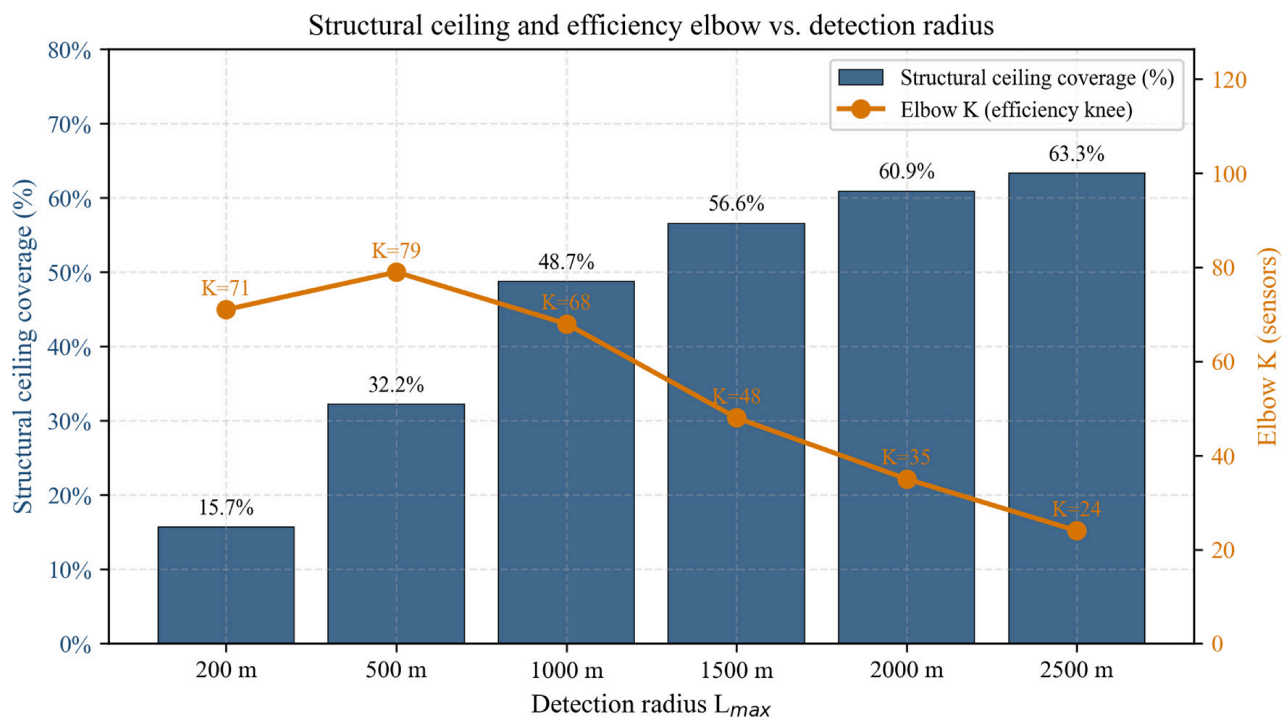


Figure 8. Structural-ceiling coverage (bars, left axis) and efficiency-elbow K (line, right axis) as a function of detection radius. The ceiling rises monotonically with radius but is bounded by isolated-fragment connectivity; the elbow shifts leftward at longer radii (after a slight rise between 200 and 500 m), confirming that extended detection range substitutes for additional sensor units.

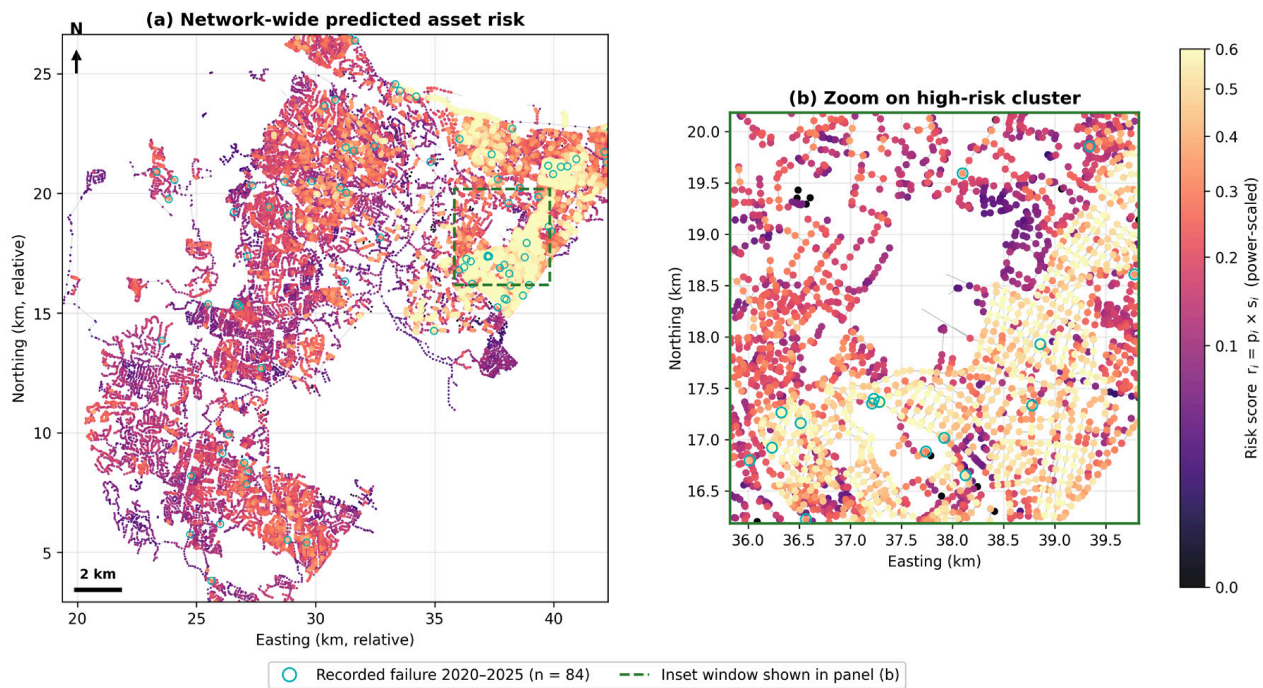


Figure 9. Predicted pipe-level risk across the anonymized case-study sewer network: (a) network-wide predicted asset risk $r_i = p_i \cdot s_i$ with recorded 2020–2025 failure locations; and (b) zoomed view of a high-risk cluster. Utility name, geographic coordinates, basemap, and other identifying information are withheld in accordance with data confidentiality requirements.

3.4. Model Interpretation and Feature Importance

The Random Forest failure model was interpreted with two complementary explainability methods—permutation importance (mean decrease in holdout ROC-AUC over 20 random feature shuffles) and SHAP values—to identify the physical factors that govern the predicted failure probability, and thus the risk weights used in the optimization. The two methods agree closely (Figure 10). Pipe age is the single dominant predictor, as permuting it reduces holdout ROC-AUC by 0.103 on average, roughly twice the effect of the next feature, pipe length (0.057), with material and slope forming a secondary tier, while diameter, pipe class, and the two graph-derived node-degree features contribute negligibly (mean permutation importance ≤ 0.004). Diameter is only a weak predictor of failure probability here, but as defined in Section 2.4, it is the primary determinant of the consequence-severity weight in the risk score, so its low importance for the probability model does not diminish its role in the overall risk. SHAP corroborates the same ranking—age first, then length—though it separates the top two features less sharply than permutation importance (mean $|\text{SHAP}| = 0.111$ for age, 0.103 for length, 0.070 for slope, and 0.050 for material). This ranking is physically consistent with gravity-sewer deterioration mechanisms—older and longer pipes accumulate more structural defects and vulnerable joints, and the material and grade further modulate the deterioration rate. The agreement of three attribution methods with different statistical assumptions (impurity-based importance, permutation importance, and SHAP) increases confidence that the risk surface reflects genuine deterioration drivers rather than an artifact of any single method, supporting the use of the predicted failure probability as the asset weight in the maximum-coverage objective. It also shows that risk weights are governed by physical attributes that are independent of the graph reconstruction, since the topology-dependent degree features are among the least important predictors.

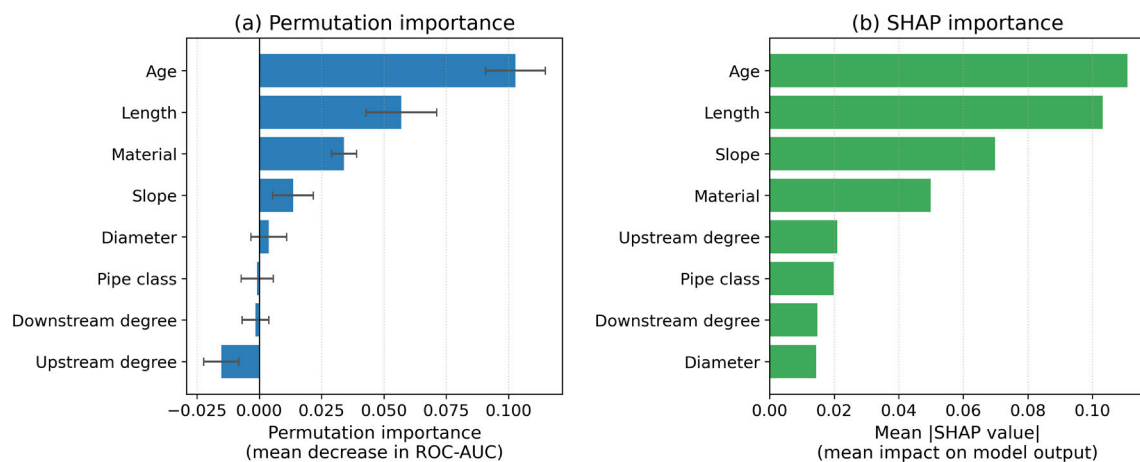


Figure 10. Explainability of the Random Forest failure model. (a) Permutation importance (mean decrease in ROC-AUC over 20 random shuffles of each feature on the holdout set; error bars show one standard deviation). (b) SHAP importance (mean absolute SHAP value across held-out pipes). Both methods identify pipe age and length as the dominant failure predictors, with material and slope secondary and the graph-derived node-degree features least important.

4. Discussion

GIS topology as a prerequisite for coverage analysis. The 32.26% optimized coverage is only achievable thanks to the GIS topology rebuild described in Section 2.1. In the raw extract the network contained 130 disconnected components and the largest covered only 29.8% of all nodes. Reverse-Dijkstra traversals were cut at every connectivity gap, so both coverage estimates and sensor placements based on the raw network were structurally invalid. The rebuild merged the network into a single 89.4% giant component and unlocked the full upstream path structure. We take this as evidence that GIS data quality has to be treated as a prerequisite, not an afterthought, for any graph-based sensor placement work. Any coverage analysis run on a fragmented network will systematically understate what is achievable.

Sensitivity to the topology reconstruction. Because the reconstruction changes graph connectivity, we tested how strongly it propagates into the two downstream products of the framework: the machine-learning risk predictions and the optimized placement. Only two of the eight predictors—upstream and downstream node degree—depend on the reconstructed topology; together they account for just 8.1% of the Random Forest feature importance, and removing them entirely changes the holdout ROC-AUC by only 0.013 (0.763 to 0.775)—a shift of about 0.6 standard deviations of the cross-validated fold distribution (standard deviation ≈ 0.022) and in the direction of slightly better performance, so the failure-probability surface is effectively invariant to the reconstruction. For the placement, the greedy optimizer was re-run at $K = 48$ on a deliberately degraded graph in which all reconstruction-added connectivity (virtual-junction and vault nodes) was removed, collapsing the largest connected component from 89.4% to below 15% of nodes—an even more fragmented state than the 29.8% raw GIS extract, and thus a conservative stress test. Because the removed vault nodes also serve as candidate sensor locations, this degraded run perturbs the candidate set as well as the connectivity, so the reported change overstates rather than understates the sensitivity to connectivity alone. Despite this large change in global connectivity, and evaluated with a stand-alone re-implementation of the coverage pipeline applied identically to both graphs—one that reproduces Table 3’s reconstructed baseline (RF with greedy at $K = 48$, 32.26%) to within about one percentage point (31.4%)—risk coverage fell by less than two percentage points, to 29.7% on the degraded graph; this 1.7-percentage-point difference between the two graphs, not the small

offset from Table 3, is the quantity of interest. The reason is that coverage is governed by local upstream reachability within the 1500 m detection radius rather than by global component structure: the reconstruction repairs GIS data-entry artifacts (endpoint-identifier mismatches and unsnapped force mains) that would otherwise sever physically continuous sewer runs, but it does not materially inflate the reported coverage. The framework's conclusions, both the machine-learning ranking and the placement coverage—are therefore robust to the reconstruction, which functions as a data-quality correction rather than a result-altering modeling choice.

The structural observability ceiling. After the rebuild, 77 isolated components remain. Together they contain about 10.6% of the nodes. These fragments hold no candidate sensors, so no sensor placed in the main network can reach an asset inside them, no matter how large the detection radius. At the same time, many pipes in long residential laterals sit beyond 4921 ft upstream of every candidate. Taken together these two constraints produce a ceiling of 56.58%, leaving 43.42% of the risk-weighted assets permanently unobservable with the current infrastructure. This 56.58% value is specific to the present candidate set and the 1500 m detection radius rather than a universal limit; because the ceiling is set jointly by which sub-basins contain candidate locations and by how far upstream each sensor can reach, it rises with detection radius (from 48.72% at 1000 m to 63.31% at 2500 m; Table 5) and would increase further if the candidate set were extended into the currently unmonitored isolated components, so it should be read as a property of the current monitoring design rather than of the sewer network alone. The $K = 400$ run confirms this empirically, as only 0.24 percentage points are added between $K = 350$ and $K = 400$. Selecting all 680 candidate locations raises coverage only to 56.60%, confirming that the 56.58% plateau is set by the candidate set and the detection radius rather than by the $K \leq 400$ search range. The ceiling is not an algorithmic shortfall. It is an actionable infrastructure finding crossing it requires expanding the candidate set into the isolated sub-networks, which is a capital-planning decision rather than a sensor-procurement one.

Table 5. Detection-radius sensitivity with the greedy algorithm, RF risk scores, and $K = 1$ to 400. Em dashes (—) indicate that the corresponding coverage milestone (40% or 50%) was not reached within the $K = 1$ to 400 search range at that detection radius. ‘Avg assets/sensor’ is the mean upstream reachable-set size averaged over all 680 candidate locations; because the greedy optimizer selects the highest-coverage candidates, the chosen $K = 48$ sensors reach far more than this pool average (8187 assets, about 171 per sensor, at the 1500 m baseline). ‘Ceiling’ is the coverage at the top of the search range ($K = 400$), which the greedy curves approach asymptotically; at the 1500 m baseline, selecting all 680 candidates yields 56.60%.

Radius L_max	Avg Assets/ Sensor	Coverage K = 48	Elbow K	Coverage K = 300	Coverage K = 400	Ceiling	K to 40%	K to 50%
200 m	10	4.22%	71	13.87%	15.71%	15.71%	—	—
500 m	23	10.81%	79	29.74%	32.20%	32.20%	—	—
1000 m	46	22.36%	68	46.97%	48.72%	48.72%	159	—
1500 m (baseline)	65	32.26%	48	55.72%	56.58%	56.58%	78	156
2000 m	80	39.09%	35	60.56%	60.91%	60.91%	51	95
2500 m	91	43.30%	24	63.14%	63.31%	63.31%	41	72

Why $K = 48$ is the practical operating point. The greedy curve shows rapid early gains, a clear efficiency elbow around $K = 48$ and sharply diminishing returns afterwards. The existing sensor budget therefore happens to sit at the natural efficiency boundary of the

current network. Past $K = 78$, the per-sensor gain has already dropped to about two-thirds of its marginal value at $K = 48$, and past $K = 156$, each new sensor contributes less than 0.4 percentage points. Practitioners can read Table 4 directly: it converts a proposed budget into an expected coverage value and shows where cost per percentage point starts to outweigh the benefit.

Detection radius and the 1500 m baseline. Across the tested range, coverage at $K = 48$ changes by a factor of ten with $L_{\max}$, which, over the ranges examined here, makes detection range a more influential parameter than either ML model or optimizer choice; this conclusion is conditional on the tested parameter ranges rather than absolute. We adopted 1500 m as the baseline for three reasons. First, it is the smallest radius at which the elbow K coincides with the actual sensor budget. Second, the 40% and 50% coverage milestones are both reached before $K = 300$. Third, while the ceiling at 1500 m (56.58%) is clearly higher than at 1000 m (48.72%), the additional gain from moving to 2000 m or 2500 m does not change any operational conclusion.

Engineering interpretation of the detection radius. Physically, the detection radius represents the along-pipe upstream distance over which a single in-network monitor can reliably register a disturbance—an anomalous level rise or flow change from a blockage or surcharge, an infiltration surge, or a water-quality excursion—originating in the sewer shed above it. It is therefore a network distance measured along the gravity flow path, not a straight-line radius, and its realistic magnitude is set by the sensing technology and by local hydraulics rather than being a free parameter. Level- and flow-based instruments, which infer upstream conditions from backwater and in-pipe storage effects, can plausibly reach the 1000–1500 m range on well-graded trunk sewers, whereas acoustic and water-quality methods, whose signals attenuate rapidly and dissipate at junctions, are typically confined to the shorter 200–500 m end of the swept range; the 2000–2500 m values represent an optimistic upper bound attainable only on steep, high-velocity mains with little branching. Hydraulic conditions modulate this range directly: in sewers meeting the minimum self-cleansing velocity of about 0.6 m/s (2 ft/s) [37], a disturbance propagates farther and more coherently, extending the effective radius, whereas flat, low-velocity laterals with frequent junctions shorten it because the signal disperses into storage and splits at each confluence. Wet-weather surcharge temporarily lengthens the detectable distance and dry-weather low flow shortens it, so the single fixed radius adopted here is best interpreted as a spatially averaged design value rather than a physical constant. For practical deployment this reframes Table 5 as a design-selection chart: a utility can enter it with the rated or field-validated detection distance of its chosen sensor class and read off the risk coverage it should realistically expect at a given budget—about 11% at $K = 48$ for a conservative 500 m level-sensing range, rising to 32% at 1500 m—rather than assuming the baseline value applies to every technology. The leftward shift in the efficiency elbow with radius carries a direct procurement message, where investment in longer-range or higher-sensitivity sensing hardware is substitutable, up to a point, for purchasing additional units; however, beyond the structural ceiling, neither better sensors nor more of them raise coverage, and only extending the monitored network into the isolated sub-basins can do so.

Optimizer and ML model choice. All three optimizers converge with identical solutions for every ML model. At $K = 48$ with 680 candidates, the problem is small enough that neither metaheuristic finds any swap-based improvement on the greedy solution, so the trivial greedy run, completing in under one second, is enough for production deployment. ML model choice does matter, as RF beats XGBoost by 0.56 percentage points and NN by 1.64 percentage points because it produces a better-calibrated risk ranking. One implication is that ranking-quality metrics (ROC-AUC and PR-AUC) are more aligned with the sensor-placement objective than a threshold-based classification metric, such

as F1—the NN attains the highest F1 yet the lowest coverage because the optimization consumes the continuous risk ranking rather than hard class labels. F1 remains a useful summary of threshold performance, but ROC-AUC and PR-AUC are the more appropriate model-selection criteria for this ranking-driven task.

Positioning relative to prior work. The framework differs from the closest prior studies in three concrete ways. First, relative to sewer sensor-placement work that optimizes uniform or information-theoretic coverage on single or synthetic networks—for example, the five formulations Banik et al. [10,11] compared on a 1902-node Italian system, the NSGA-II placement Wang et al. [12] applied to a 4394-node Hong Kong network, and the spectral-clustering and joint-entropy approach of Wu et al. [13]—the present study weights coverage by an ML-predicted, per-pipe failure-risk score and evaluates it on the real, 33,349-pipe layout of an operating U.S. utility. Second, relative to the sewer failure-prediction literature that terminates at a trained classifier [25–27,33], this study carries the predicted risk forward into a prescriptive placement decision, closing the predictive-to-prescriptive loop end to end. Third, relative to the water-distribution literature, where placement objectives are weighted by the probability and consequence of contamination events [7] (as surveyed by Rathi et al. [36]), this study weights coverage by a machine-learning-predicted per-pipe failure likelihood combined with a consequence severity weight, transferring the principle to gravity sewer collection systems and to a physical-failure-risk objective, and adds three elements that prior sewer studies do not report together: a GIS topology reconstruction that is a prerequisite for meaningful upstream-coverage analysis, a fixed-denominator coverage definition that makes results from different ML models directly comparable, and an explicit quantification of the structural observability ceiling imposed by residual network fragmentation. Finally, the optimized placements are benchmarked against the utility’s actual 48-sensor field deployment rather than a synthetic baseline.

Comparison with literature. The results align with, and extend, several findings in the prior literature. On the predictive side, the Random Forest holdout ROC-AUC of 0.763 sits at the upper end of the 0.60–0.68 range that Aounali et al. [33] reported as typical for static-attribute sewer models, below the 0.81–0.90 they obtained after adding dynamic hydraulic features; therefore, our result confirms that a well-engineered static-feature model is competitive and identifies dynamic-feature enrichment as the most promising route to higher accuracy, consistent with [33]. On the prescriptive side, three observations echo established results. First, the finding that greedy is effectively optimal—the genetic algorithm and tabu search converge to the greedy solution to within 0.01 percentage points—independently reproduces the submodular near-optimality of greedy reported by Krause et al. [6] and the conclusion of Banik et al. [10,11] that greedy offers the best quality–runtime trade-off among five formulations on a real sewer network. Second, the magnitude of the improvement from optimized placement (a near-tripling of risk coverage, from 11.73% to 32.26%, a 174.9% gain over the existing layout) is comparable in scale to the 102% improvement Wang et al. [12] reported for NSGA-II over centrality heuristics and the 7–15% gains Wu et al. [13] reported over NSGA-III, while being achieved with a simpler, near-optimal greedy solver. Third, and in contrast to these prior studies—which optimize uniform or information-theoretic coverage on single or synthetic networks—our gains are measured on a real utility layout under a failure-risk-weighted objective, so the absolute percentages are not directly comparable, but the qualitative agreement (greedy near-optimality and large gains over unoptimized baselines) strengthens confidence that the risk-weighted formulation carries the benefits documented in the water-distribution and idealized-sewer literature into an operational sewer setting.

Implementation pathway for practitioners. The framework is designed to run on data that utilities already maintain, so adoption does not require new field campaigns. In practice a utility would follow four steps: (i) Assemble inputs from existing systems, the GIS geodatabase for pipe geometry and attributes, and maintenance, CCTV-inspection, and overflow records for the failure labels. (ii) Run the descriptive–predictive–prescriptive pipeline to produce a per-pipe risk surface and a ranked list of recommended sensor locations. (iii) Act on the ranked list at whatever budget can be funded as an immediate step requiring no additional sensor purchases, relocate existing meters toward the top-ranked sites (which nearly triples risk coverage in the case study), and, for planned capital, use the coverage-versus-budget curve (Table 4 and Figure 5) to set defensible cost-coverage targets and to justify sensor purchases to decision-makers. (iv) Fold the model into the asset-management cycle, re-running it as new inspection and failure data accrue. The expected impact follows directly from the objective: by concentrating limited monitoring on the highest-risk, most-observable pipes, a utility raises the probability of detecting a developing failure before it becomes a sanitary sewer overflow, thereby reducing environmental releases, regulatory penalties, and costly emergency repairs, and producing the risk-based inspection targeting that supports proactive rehabilitation planning.

Software and skills requirements. The entire pipeline is implemented in open-source Python using standard, freely available libraries (scikit-learn 1.4.2 for the classifiers, NetworkX 3.2.1 for graph construction and reverse-Dijkstra reachability, and pandas 2.2.1/GeoPandas 0.14.4 for data handling); no proprietary or specialized modeling software is required, and a full production run—GIS reconstruction, Random Forest training, and greedy placement at the baseline detection radius—completes in under two minutes on a standard workstation; the genetic algorithm and tabu search optimizers reported in Table 3 are slower (tabu search up to about two minutes per model) but serve only to confirm that no swap-based improvement on the greedy solution exists and are not required for deployment. We nonetheless recognize that not every utility maintains in-house data-science capacity, and the workflow is deliberately designed so that building the model and using its outputs are separable tasks. The one-time model build—feature assembly, training, and optimization—can be carried out by a utility data analyst, an academic partner, or a consultant, after which re-running the pipeline on updated data is effectively a push-button process. The outputs that practitioners act on are intentionally simple: a ranked table of recommended sensor locations that can be joined back to the utility’s GIS as points and reviewed by a GIS technician without any machine-learning expertise. Because most utilities already operate a GIS platform that supports Python-based geoprocessing, the pipeline can also be packaged as a GIS toolbox or a lightweight script so that staff run it through a familiar interface.

Applying the framework to new data or other utilities. Because every stage of the pipeline is driven by the input files rather than by network-specific tuning, applying it to an updated dataset or to a different utility requires only that the new data be supplied in the same schema. In practice, the user exports a current GIS extract (pipe geometry and attributes) and the latest failure records (inspection, maintenance, and overflow logs); the topology-rebuild stage then reconstructs the graph, the eight features are recomputed, the classifier is refitted—or the saved model applied—to produce an updated per-pipe risk surface, and the optimizer returns a refreshed ranked list of sensor locations. For an existing deployment, this is the recommended annual, or event-triggered, retraining cycle, which keeps the risk surface aligned with the most recent condition data and completes in under two minutes. For a different utility, the same procedure applies unchanged; only the absolute coverage values, the ceiling, and the selected sites are network-specific, whereas the methodology and the code transfer directly, as noted in the Limitations. The

pipeline also degrades gracefully when a new dataset is less complete than the one used here: missing attributes are median-imputed and ages are clipped exactly as described in Section 2.3, so a utility with a partially populated GIS can still obtain a usable risk ranking, albeit with correspondingly wider uncertainty.

Practical recommendations. The highest-return immediate action for the utility is to redeploy the existing 48 sensors to RF-greedy optimal locations. Of the 48 existing stations, 12 already coincide with optimal locations, so this redeployment repositions the remaining 36; it requires no new hardware and lifts risk coverage from 11.73% to 32.26%, an increase of 174.9%. This existing 11.73% coverage is not a failed risk design but the result of a different original objective—these stations were situated for hydraulic monitoring (flow accounting and wet-well control) rather than risk surveillance—so the improvement reflects re-purposing rather than the correction of a design flaw. Because some meters also serve billing or pump-station control and may not be freely relocatable, a partial-relocation variant that moves only the unconstrained subset is the realistic near-term action. For incremental budget growth, the efficiency elbow points to $K = 78$ as the next milestone (30 extra sensors for an extra 7.86 percentage points), and $K = 156$ to cross the 50% threshold. Beyond $K = 300$ or so, additional spending is better directed at extending the candidate set into the isolated sub-networks than at buying more sensors. The RF model should be retrained once a year as new failure records come in; re-running the entire placement optimization finishes in under a minute. Regional cost-estimation guidance for any companion pipe-renewal investments, such as the trenchless rehabilitation work that typically follows a risk-driven inspection campaign, is provided by Hamidzadeh et al. [42].

Limitations. Several limitations should be noted. The absolute coverage numbers, the ceiling value, the elbow position and the selected sites are specific to the network studied and to the 2020 to 2025 data vintage; the methodology transfers directly to other utilities, but the absolute numbers will not. The failure labels were assigned by spatial join, which leaves about 1 to 5 m of positional uncertainty, and a random-sample CCTV check of labels would strengthen confidence in the training set. The severity weight s_i currently depends only on the diameter and pipe type; richer consequence data, such as population served, proximity to sensitive receptors and repair cost, would yield a more defensible ranking. The 56.58% ceiling is a property of the specific 680-candidate set used here; a candidate set that includes manholes inside the isolated fragments would produce a different ceiling. Finally, the analysis treats the network as static, and the annual retraining cycle is recommended as a practical response to that limitation.

5. Conclusions

We have presented DPP-SP, a Descriptive–Predictive–Prescriptive Sensor-Placement framework for risk-driven monitoring of large sanitary sewer collection systems and applied it to the 33,349-pipe sewer network of a large U.S. utility. The framework combines GIS topology rebuilding, machine-learning failure prediction, and maximum weighted-coverage optimization to pick K sensor locations that maximize risk-weighted upstream detectability. At $K = 48$, the framework reaches 32.26% risk coverage, against an existing baseline of 11.73%, an improvement of 174.9%. The best results come from Random Forest risk scores with greedy placement, and all three optimizer variants (greedy, GA, TS) converge on the same solution. Extending the search to $K = 1$ to 400 uncovers a structural ceiling of 56.58%—specific to the current candidate set and 1500 m detection radius and rising with radius as the sensitivity analysis shows—caused by 77 residual isolated network fragments and the 4921 ft detection-distance constraint. Within the ranges tested, a six-radius sensitivity study identifies detection range as more influential on coverage than either the ML model or the optimizer choice. From a practical standpoint,

the main implication is that the utility can nearly triple its risk coverage without purchasing additional sensors by moving its 48 sensors to the optimized sites. From a methodological standpoint, the main implication is that risk-weighted, ML-informed coverage objectives produce noticeably better sensor allocations than the uniform-coverage formulations that still dominate the sewer-placement literature.

DPP-SP is deliberately modular: the GIS-rebuild, ML-risk and placement stages can each be replaced independently. We invite the community to extend the framework along six open directions. (i) Candidate-set expansion. Enlarging the candidate pool into the currently isolated network fragments would break the 56.58% structural ceiling; this is a capital-planning problem well-suited to multi-criteria decision analysis. (ii) Adaptive re-placement. Feeding real-time sensor telemetry back into the failure model would create a closed-loop digital twin that continuously updates p_i and re-optimizes sensor locations as new readings arrive. (iii) Multi-objective formulation. Extending the single-objective coverage maximization to a Pareto formulation that jointly balances risk coverage, installation cost, maintenance accessibility and equity of service across sub-watersheds would yield more defensible deployment plans. (iv) Cross-utility benchmarking. The fixed-denominator coverage definition introduced here is designed for cross-network comparison; applying DPP-SP to additional utilities of different size, climate and topology would produce the first benchmark suite for sewer sensor placement. (v) Graph-learning extension. Replacing the tabular Random Forest with a graph neural network that learns directly on the sewer graph could exploit connectivity features the current eight-feature vector ignores. (vi) Coupling with hydraulic simulation. Joint optimization of sensor placement and SWMM-based inflow/infiltration detectability would close the loop between condition monitoring and real-time control. We hope these directions will accelerate the adoption of risk-weighted, ML-informed sensor placement across the wastewater industry.

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Data Availability Statement: The GIS network, failure records, monitoring locations, and sensor telemetry analyzed in this study were provided by the participating utility under a non-disclosure agreement and cannot be publicly redistributed. The agreement identifies utility-provided materials, including maps, research results, designs, data, and related information, as confidential or proprietary information, and further disclosure requires written authorization from the utility. Therefore, raw utility data and utility-specific derived outputs, including asset identifiers, geospatial files, pipe-level risk scores, failure locations, candidate monitoring locations, and optimized sensor configurations, are not publicly available. To support reproducibility and follow-on research, the DPP-SP reference implementation will be released as an open-source Python package upon publication. The package will include topology-rebuilding routines; machine-learning training scripts; coverage-set construction; Dijkstra-based upstream tracing; greedy, genetic algorithm, and tabu search optimization routines; and the radius-sensitivity analysis workflow. Reproducible example workflows and outputs will be provided using anonymized or synthetic data only. Any request for access to utility-specific materials would require written authorization from the participating utility and must comply with the applicable data-use and confidentiality agreement.

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References

1. EPA. *Report to Congress on Impacts and Control of Combined Sewer Overflows and Sanitary Sewer Overflows*; EPA/833-R-04-001; U.S. Environmental Protection Agency: Washington, DC, USA, 2004.
2. ASCE. *2021 Report Card for America's Infrastructure: Wastewater*; American Society of Civil Engineers: Reston, VA, USA, 2021.
3. Calderon, D.R.; Moradi, M.; Najafi, M.; Kaushal, V.; Habibian, A.; Villar, G. Insights on condition monitoring of water assets from the results of an industry survey. In Proceedings of the Pipelines 2025 Conference, ASCE, Reston VA, USA, 9–13 August 2025. [\[CrossRef\]](#)
4. Moradi, M.; Najafi, M.; Lin, Z.; Pan, H. Advancing sensor network optimization and data analytics for smart wastewater and stormwater collection systems: An overview. In Proceedings of the Pipelines 2025 Conference, ASCE, Reston, VA, USA, 9–13 August 2025. [\[CrossRef\]](#)
5. Lepenioti, K.; Bousdekis, A.; Apostolou, D.; Mentzas, G. Prescriptive analytics: Literature review and research challenges. *Int. J. Inf. Manag.* **2020**, *50*, 57–70. [\[CrossRef\]](#)
6. Krause, A.; Leskovec, J.; Guestrin, C.; VanBriesen, J.; Faloutsos, C. Efficient sensor placement optimization for securing large water distribution networks. *J. Water Resour. Plan. Manag.* **2008**, *134*, 516–526. [\[CrossRef\]](#)
7. Leskovec, J.; Krause, A.; Guestrin, C.; Faloutsos, C.; VanBriesen, J.; Glance, N. Cost-effective outbreak detection in networks. In Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Jose, CA, USA, 12–15 August 2007; pp. 420–429.
8. Alfonso, L.; Lobbrecht, A.; Price, R. Optimization of water level monitoring network in polder systems using information theory. *Water Resour. Res.* **2010**, *46*, W12553. [\[CrossRef\]](#)
9. Berry, J.; Hart, W.E.; Phillips, C.A.; Uber, J.G.; Watson, J.-P. Sensor placement in municipal water networks with temporal integer programming models. *J. Water Resour. Plan. Manag.* **2006**, *132*, 218–224. [\[CrossRef\]](#)
10. Banik, B.K.; Alfonso, L.; Di Cristo, C.; Leopardi, A. Greedy algorithms for sensor location in sewer systems. *Water* **2017**, *9*, 856. [\[CrossRef\]](#)
11. Banik, B.K.; Alfonso, L.; Di Cristo, C.; Leopardi, A.; Mynett, A. Evaluation of different formulations to optimally locate sensors in sewer systems. *J. Water Resour. Plan. Manag.* **2017**, *143*, 04017026. [\[CrossRef\]](#)
12. Wang, X.; Xu, P.; Zhou, Y. Optimal placement of monitoring sensors in a large-scale sewage network using multi-objective optimization. *Water Res.* **2024**, *257*, 121684.
13. Wu, Y.; Zheng, F.; Yang, Y.; Zhang, K.; Du, K.; Duan, H.; Savic, D.; Kapelan, Z. Enhancing inflow and infiltration detection in urban sewer networks with a new deterministic sensor placement method. *Water Res.* **2025**, *268*, 122591. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Nourinejad, M.; Berman, O.; Larson, R.C. Placement of sensors in sewersheds for wastewater-based epidemiology. *Eur. J. Oper. Res.* **2021**, *295*, 1305–1316.
15. Ding, X.; Zhang, X. Sparse sensing for stormwater monitoring via SVD and QR factorization. *J. Hydrol.* **2025**, *632*, 130892.
16. Kerkez, B.; Gruden, C.; Lewis, M.; Montestruque, L.; Quigley, M.; Wong, B.; Bedig, A.; Kertesz, R.; Braun, T.; Cadwalader, O.; et al. Smarter stormwater systems. *Environ. Sci. Technol.* **2016**, *50*, 7267–7273. [\[CrossRef\]](#) [\[PubMed\]](#)
17. Mullapudi, A.; Lewis, M.J.; Gruden, C.L.; Kerkez, B. Deep reinforcement learning for the real-time control of urban drainage systems. *Adv. Water Resour.* **2020**, *140*, 103600. [\[CrossRef\]](#)
18. Bartos, M.; Kerkez, B. Pipedream: An interactive digital twin model for natural and urban drainage systems. *Environ. Model. Softw.* **2021**, *144*, 105120. [\[CrossRef\]](#)
19. Bakhshipour, A.E.; Hespen, J.; Haghighi, A.; Dittmer, U.; Nowak, W. Integrating structural resilience in the design of urban drainage networks in flat areas using a simplified multi-objective optimization framework. *Water* **2021**, *13*, 269. [\[CrossRef\]](#)
20. Breiman, L. Random forests. *Mach. Learn.* **2001**, *45*, 5–32. [\[CrossRef\]](#)
21. Mohammadi, M.M.; Najafi, M.; Kaushal, V.; Serajiantehrani, R.; Salehabadi, N.; Ashoori, T. Sewer pipes condition prediction models: A state-of-the-art review. *Infrastructures* **2019**, *4*, 64. [\[CrossRef\]](#)
22. Hawari, A.; Alkadour, F.; Elmasry, M.; Zayed, T. A state-of-the-art review on condition assessment models developed for sewer pipelines. *Eng. Appl. Artif. Intell.* **2020**, *93*, 103721. [\[CrossRef\]](#)
23. Tscheikner-Gratl, F.; Caradot, N.; Cherqui, F.; Leitão, J.P.; Ahmadi, M.; Langeveld, J.G.; Le Gat, Y.; Scholten, L.; Roghani, B.; Rodríguez, J.P.; et al. Sewer asset management—State of the art and research needs. *Urban Water J.* **2019**, *16*, 662–675. [\[CrossRef\]](#)

24. Salihu, C.; Hussein, M.; Mohandes, S.R.; Zayed, T. Towards a comprehensive review of the deterioration factors and modeling for sewer pipelines: A hybrid of bibliometric, scientometric, and meta-analysis approach. *J. Clean. Prod.* **2022**, *351*, 131460. [\[CrossRef\]](#)
25. Laakso, T.; Kokkonen, T.; Mellin, I.; Vahala, R. Sewer condition prediction and analysis of explanatory factors. *Water* **2018**, *10*, 1239. [\[CrossRef\]](#)
26. Tavakoli, R.; Sharifara, A.; Najafi, M. Prediction of pipe failures in wastewater networks using random forest. *J. Pipeline Syst. Eng. Pract.* **2020**, *11*, 04020041.
27. Hassouna, M.; Tarhini, M.; Jamal, D. A machine-learning approach for sewer-pipe condition prediction. *Urban Water J.* **2019**, *16*, 89–99.
28. Chen, T.; Guestrin, C. XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, 13–17 August 2016; pp. 785–794.
29. Robles-Velasco, A.; Ramos-Salgado, C.; Muñuzuri, J.; Cortés, P. Artificial neural networks to forecast failures in water supply pipes. *Sustainability* **2021**, *13*, 8226. [\[CrossRef\]](#)
30. Winkler, D.; Haltmeier, M.; Kleidorfer, M.; Rauch, W.; Tscheikner-Gratl, F. Pipe failure modelling for water distribution network using boosted decision trees. *Struct. Infrastruct. Eng.* **2018**, *14*, 1402–1411. [\[CrossRef\]](#)
31. Fontecha, J.E.; Agarwal, P.; Torres, M.N.; Mukherjee, S.; Walteros, J.L.; Rodríguez, J.P. A two-stage data-driven framework for spatiotemporal analysis of urban sewer systems. *J. Clean. Prod.* **2021**, *285*, 124890.
32. Wijs, A.; Hooyberghs, T.; Vansteenkiste, L.; Mertens, E. Machine-learning approaches for predicting sewer pipe failures with imbalanced data. *Water* **2020**, *12*, 2367.
33. Aounali, O.; Shepherd, W.; Tait, S. Integrating dynamic features into machine-learning models for predicting sewer network failures: A Random Forest approach. *Urban Water J.* **2026**, *23*, 408–423. [\[CrossRef\]](#)
34. Purdy, G. ISO 31000:2009—Setting a new standard for risk management. *Risk Anal.* **2010**, *30*, 881–886. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Anbari, M.J.; Tabesh, M.; Roozbahani, A. Risk assessment model to prioritize sewer pipes inspection in wastewater collection networks. *J. Environ. Manag.* **2017**, *190*, 91–101. [\[CrossRef\]](#) [\[PubMed\]](#)
36. Rath, S.; Gupta, R.; Ormsbee, L. A review of sensor placement objective metrics for contamination detection in water distribution networks. *Water Sci. Technol. Water Supply* **2016**, *16*, 1356–1366.
37. ASCE. *Gravity Sanitary Sewer Design and Construction, Manual of Engineering Practice No. 36*, 2nd ed.; American Society of Civil Engineers: Reston, VA, USA, 2007.
38. Great Lakes–Upper Mississippi River Board. *Recommended Standards for Wastewater Facilities (Ten States Standards)*; Health Education Services: Albany, NY, USA, 2004.
39. Feige, U. A threshold of $\ln n$ for approximating set cover. *J. ACM* **1998**, *45*, 634–652. [\[CrossRef\]](#)
40. Nemhauser, G.L.; Wolsey, L.A.; Fisher, M.L. An analysis of approximations for maximizing submodular set functions—I. *Math. Program.* **1978**, *14*, 265–294. [\[CrossRef\]](#)
41. EPA. *Condition Assessment of Wastewater Collection Systems*; EPA/600/R-12/709; U.S. Environmental Protection Agency, Office of Research and Development: Washington, DC, USA, 2012.
42. Hamidzadeh, P.; Moradi, M.; Najafi, M. Developing a regionally adaptable cost estimation guide for pipeline renewal and replacement using trenchless technologies. In Proceedings of the North American Society for Trenchless Technology (NASTT) 2026 No-Dig Show, Palm Springs, CA, USA, 29 March–2 April 2026.

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